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USERS: NAVY - YD
- MC

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P.A. ARMY-ME
Other Cust
AIR FORCE-82

PROCUREMENT SPECIFICATION
MIL-S-3785

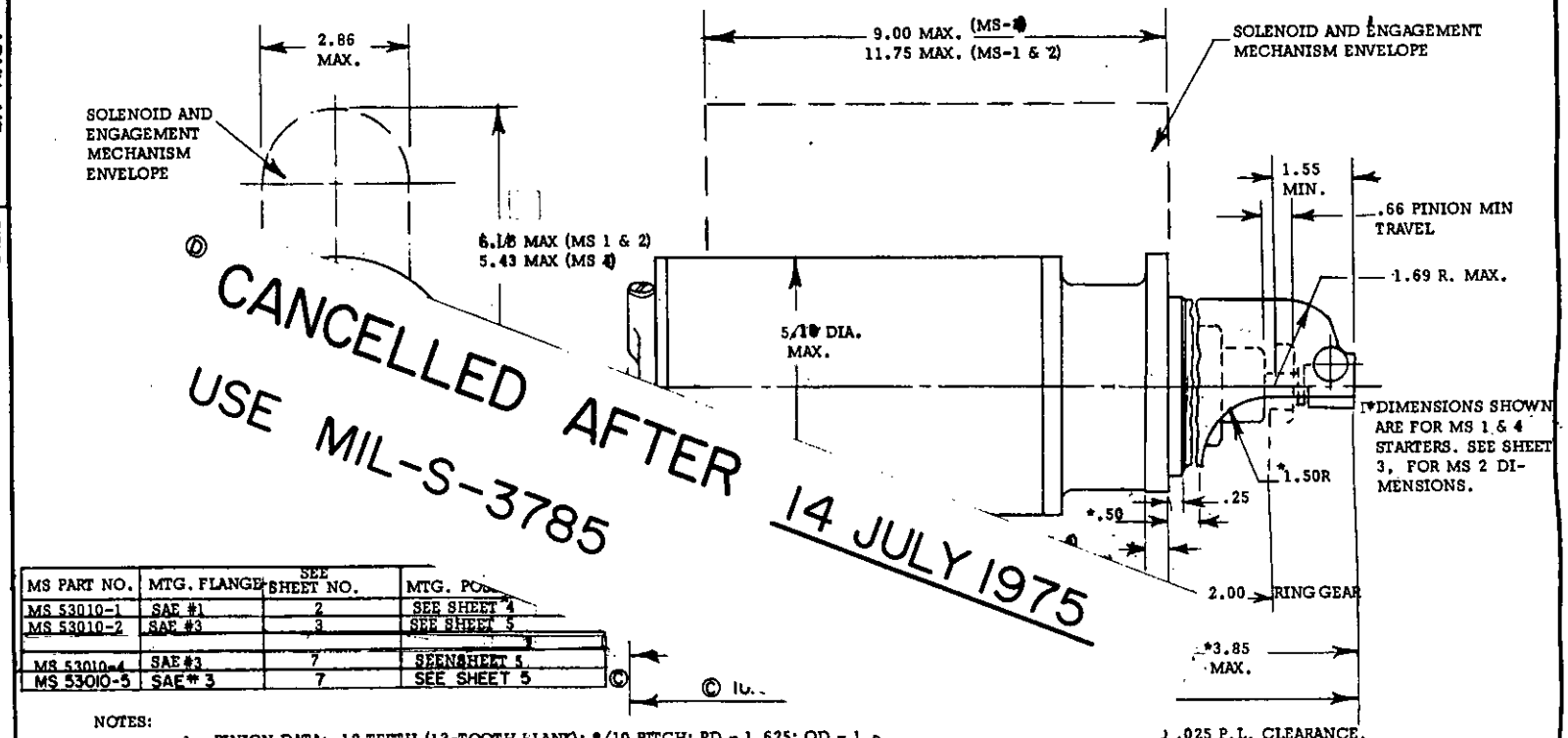
SUPERSEDES:

TITLE
STARTER, ENGINE, ELECTRICAL: 24-VOLT, D.C., 5-1/8-INCH FRAME DIAMETER, SOLENOID-ACTUATED, LIGHT-DUTY, SHORT AND LONG FRAME

MILITARY STANDARD

MS 53010

SHEET 1 OF 7



© CANCELLED AFTER 14 JULY 1975 USE MIL-S-3785

MS PART NO.	MTG. FLANGE	SEE SHEET NO.	MTG. POS
MS 53010-1	SAE #1	2	SEE SHEET 4
MS 53010-2	SAE #3	3	SEE SHEET 5
MS 53010-4	SAE #3	7	SEE SHEET 5
MS 53010-5	SAE #3	7	SEE SHEET 5

NOTES:

1. PINION DATA: 12 TEETH (13-TOOTH BLANK); 8/10 PITCH; PD - 1.625; OD - 1.6..
2. MOTOR MAY BE USED WITH EITHER NEGATIVE OR POSITIVE GROUNDED SYSTEM.
3. MOUNTING IS SIMILAR TO #1 OR #3 SAE FLANGE MOUNTING (SEE TABLE).
4. TERMINAL THREAD SIZE: BATTERY - 3/8 - 16 UNC 2A
   SWITCH - #10 - 32 UNF 2A
5. DESIGN AND ARRANGEMENT OF SOLENOID AND ENGAGEMENT MECHANISM ASSEMBLY ARE OPTIONAL, PROVIDED SHIFT LEVER AND LINKAGE ARE INCLOSED AND ALSO PROVIDED TERMINAL THREAD SIZES CONFORM TO AND LIMITING OVERALL DIMENSIONS DO NOT EXCEED THOSE SPECIFIED HEREIN.
6. ROTATION VIEWING DRIVE END.
7. TOLERANCE: DIMENSIONAL; ± .030 UNLESS OTHERWISE SPECIFIED. ANGULAR; ± 1°.
8. THIS MOTOR IS IDENTIFIED AS TYPE II (RING GEAR ENGAGEMENT), GRADE A (WATERPROOF), CLASS 1 (POSITIVE DRIVE) OF REFERENCED SPECIFICATION.
9. MOTOR SHALL BE EQUIPPED WITH OIL SEAL FOR WET CLUTCH OPERATION.
10. UNLESS OTHERWISE SPECIFIED, ALL DIMENSIONS ARE IN INCHES.
11. MODIFICATIONS IN OUTSIDE APPEARANCE AND CONSTRUCTION ARE ACCEPTABLE IF UNIT INTERCHANGEABILITY IS NOT AFFECTED.
12. FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER DOCUMENTS REFERENCED HEREIN.
13. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATION FOR BID.

(NOTES CONTINUED ON SHEET 6)

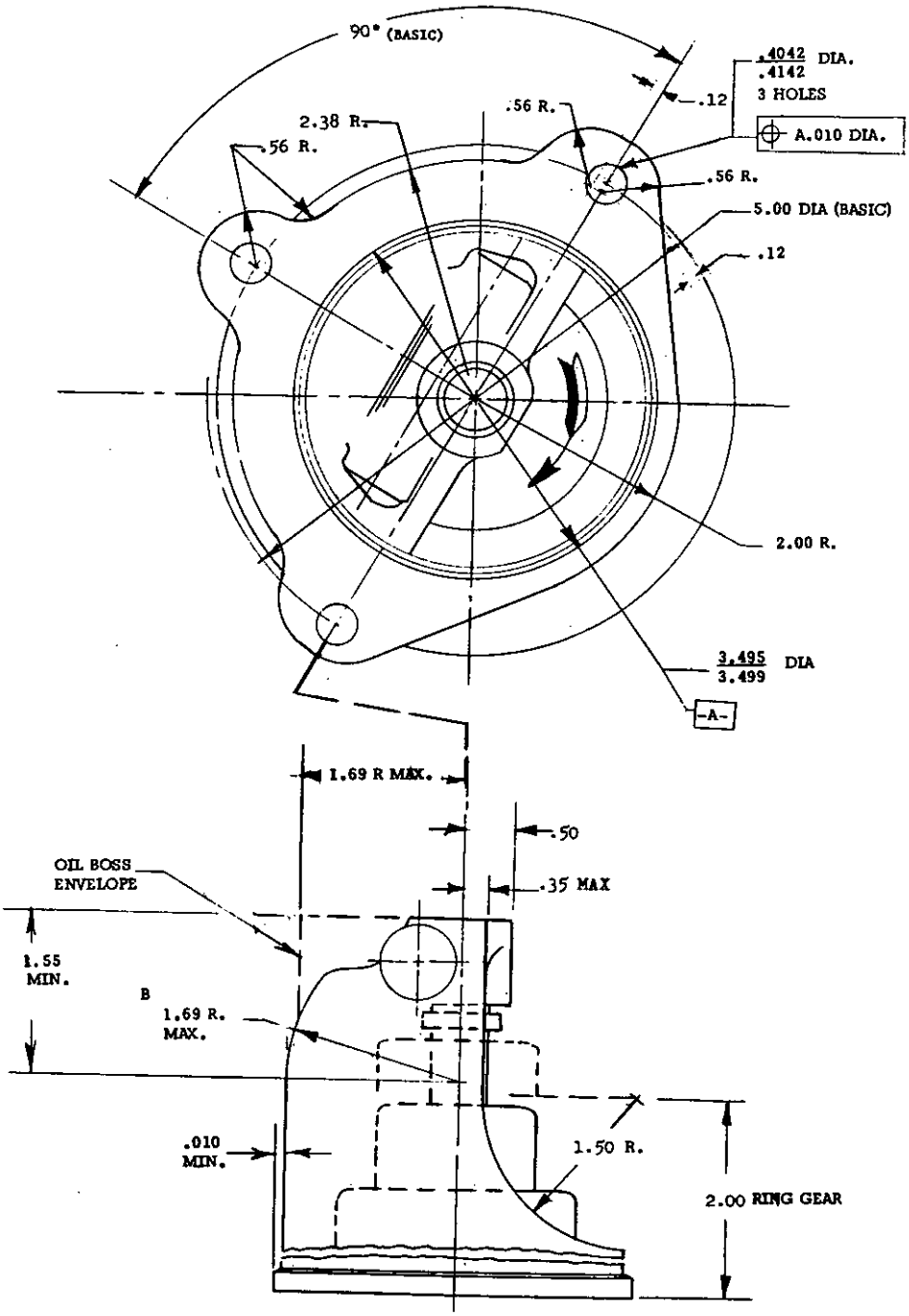
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SAE # 1 MOUNTING FLANGE (SEE NOTE 3)

FOR MS-1 STARTER

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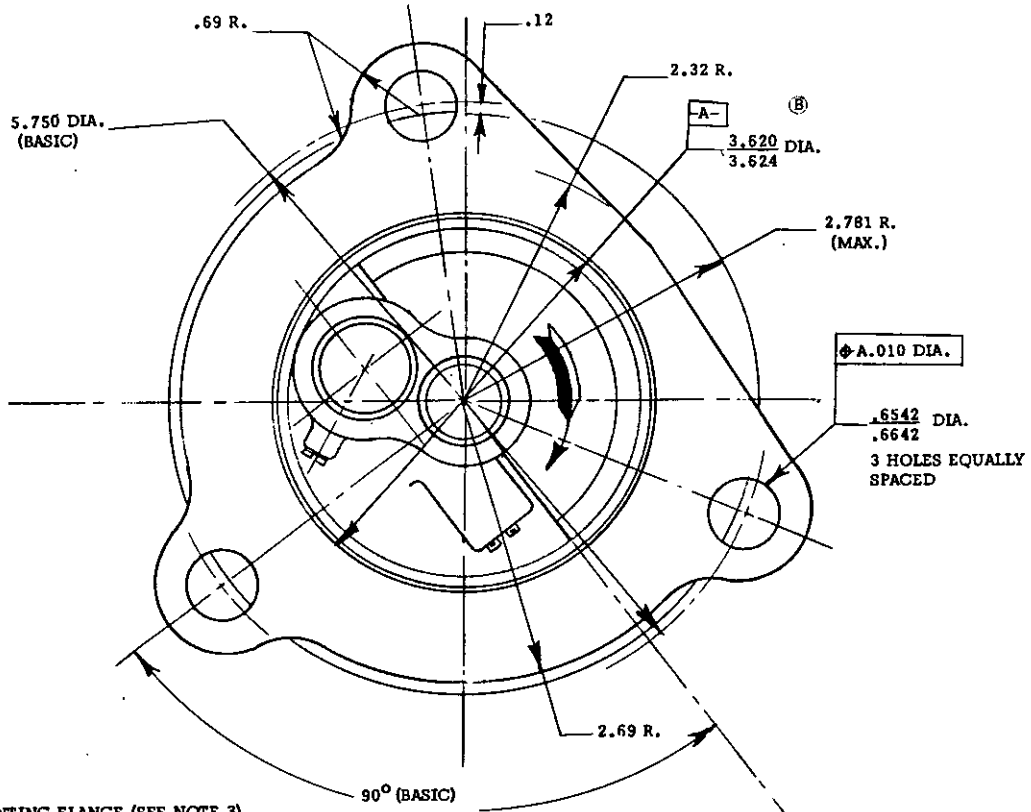
P.A. ARMY - ME Other Cust. AIR FORCE - 82	TITLE STARTER, ENGINE, ELECTRICAL, 24-VOLT, D. C., 5-1/8-INCH FRAME DIAMETER, SOLENOID-ACTUATED, LIGHT-DUTY, SHORT AND LONG FRAME	MILITARY STANDARD MS 53010
PROCUREMENT SPECIFICATION MIL-S-3785	SUPERSEDES:	SHEET 2

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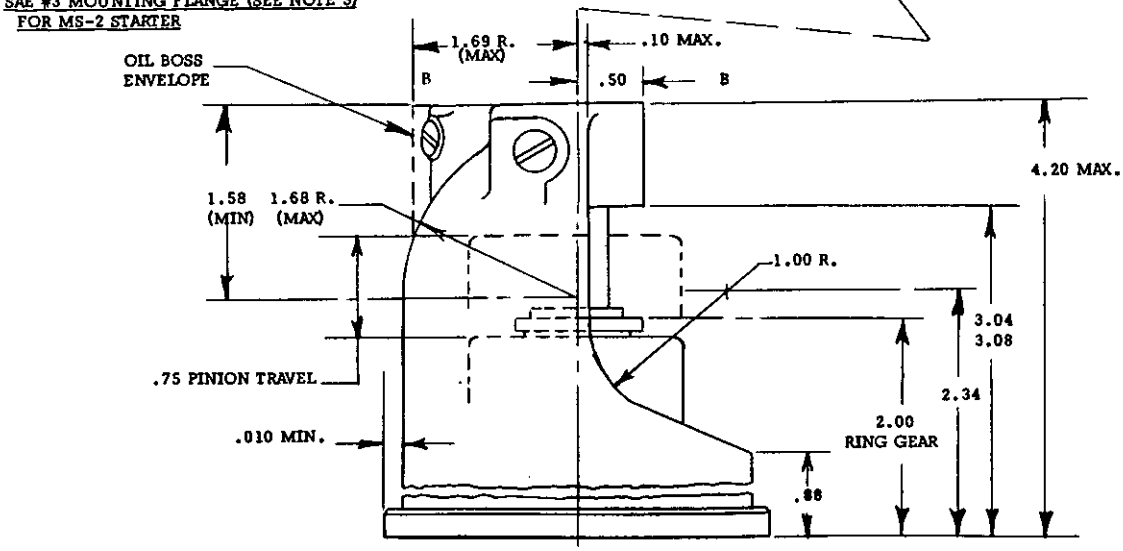
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SAE #3 MOUNTING FLANGE (SEE NOTE 3)
FOR MS-2 STARTER



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Ⓓ FOR CHANGES SEE SHEET 1

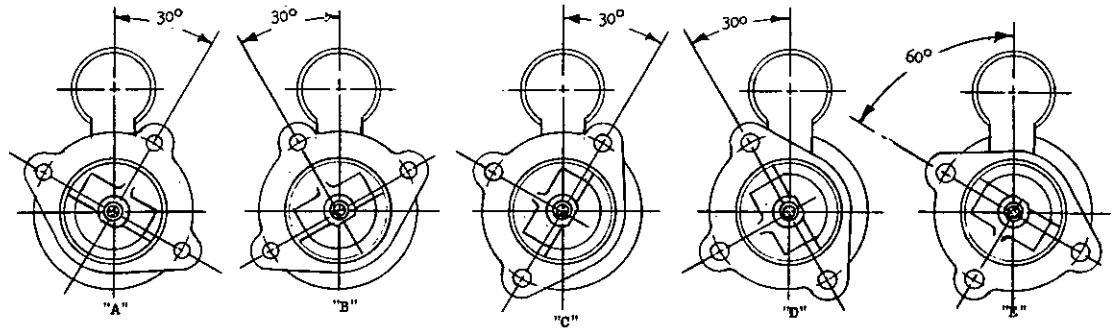
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PROCUREMENT SPECIFICATION MIL-S-3785	SUPERSEDES:	SHEET 3

DD FORM 1 SEP 57 672-1 (Coordinated)

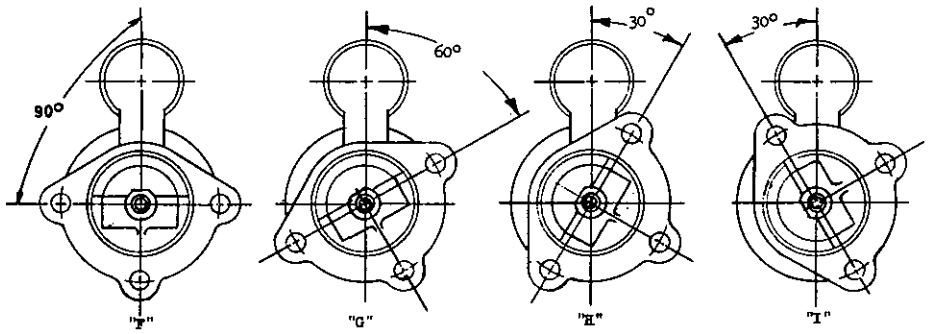
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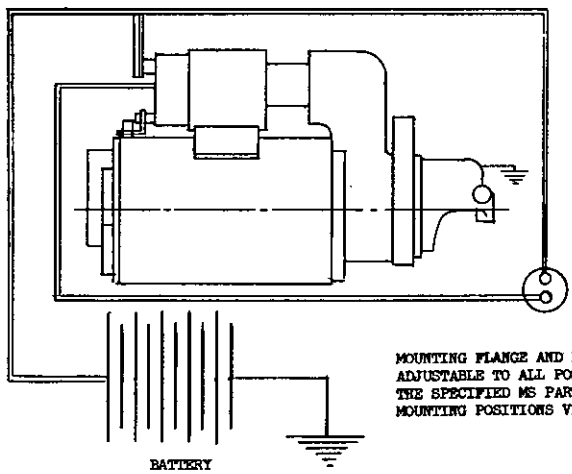
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#1 SAE MOUNTING FLANGE (SEE NOTE 3)
FOR MS-1 STARTER





MOUNTING FLANGE AND PINION HOUSING SHALL BE ADJUSTABLE TO ALL POSITIONS SHOWN HEREON FOR THE SPECIFIED MS PART NO. (SEE SHEET 1). MOUNTING POSITIONS VIEWING DRIVE END OF MOTOR.

TYPICAL WIRING DIAGRAM

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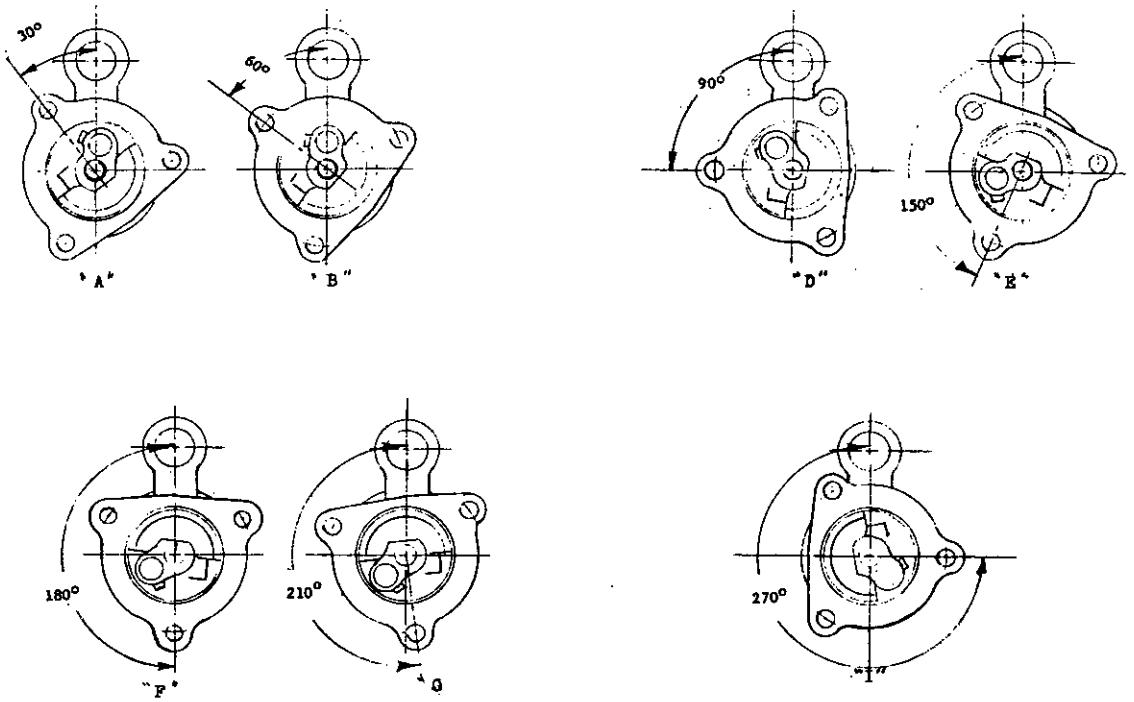
P.A. ARMY-ME Other Cust. AIR FORCE-82	TITLE STARTER ENGINE, ELECTRICAL, 24-VOLT, D. C., 5-1/8-INCH FRAME DIAMETER, SOLENOID-ACTUATED, LIGHT-DUTY, SHORT AND LONG FRAME	MILITARY STANDARD MS 53010
PROCUREMENT SPECIFICATION MIL-S-3785	SUPERSEDES:	SHEET 4

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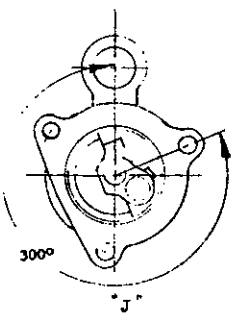
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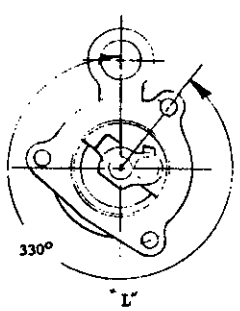


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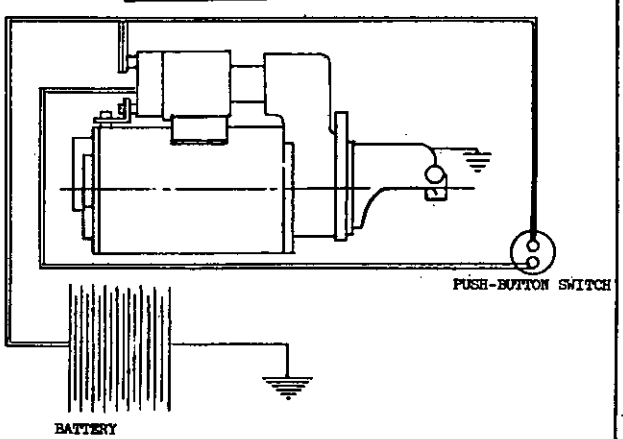
MOUNTING FLANGE AND PINION HOUSING SHALL BE ADJUSTABLE TO ALL POSITIONS SHOWN HEREON. MOUNTING POSITIONS VIEWING DRIVE END OF MOTOR.



FIGURES "C", "H", AND "K" HAVE BEEN DELETED



3 SAE MOUNTING FLANGE (SEE NOTE 3) FOR MS-2, MS-4 AND MS-5 STARTERS ©



TYPICAL WIRING DIAGRAM

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P.A. ARMY-ME
Other Cust.
AIR FORCE-82

PROCUREMENT SPECIFICATION
MIL-S-3785

MILITARY STANDARD

STARTER, ENGINE, ELECTRICAL, 24-VOLT, D.C., 5-1/8-INCH FRAME DIAMETER, SOLENOID-ACTUATED, LIGHT-DUTY, SHORT AND LONG FRAME

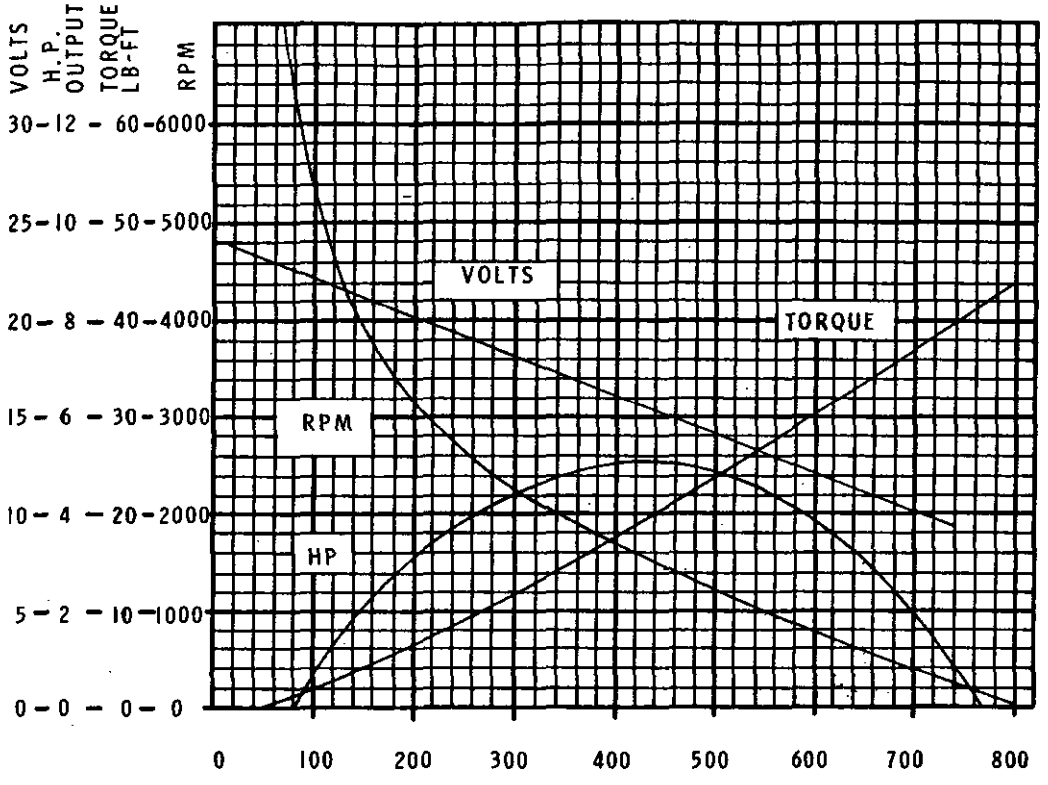
SUPERSEDES:

MS 53010

SHEET 5

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PERFORMANCE CURVE

(CONTINUATION OF NOTES FROM SHEET 1)

14. TORQUE AND R.P.M. MUST NOT BE LESS THAN 10% BELOW THAT SHOWN BY CURVES, WITH VOLTAGE APPLIED AS SPECIFIED. OUTPUT CURVE IS NOT RELATED TO TORQUE AND R.P.M. CURVES SHOWN, BUT IS THE MINIMUM FOR ANY PARTICULAR MOTOR.
15. THE VOLTAGE CURVE SHOWN IS THE SAE STANDARD DUTY 24-VOLT BATTERY CURVE.
16. LOCATION OF NAMEPLATE SHALL BE OPTIONAL. NAMEPLATE SHALL BE FASTENED TO STARTER FRAME WITH RIVETS OR SCREWS AND LOCK WASHERS.
17. STALLED TORQUE DATA:

    VOLTS - 5.1 (APPROX)
    CURRENT (AMPS) (MAX) - 500
    TORQUE (LB-FT) (MIN) - 18

18. SOLENOID CURRENT CONSUMPTION (PULL-IN PLUS HOLD-IN) SHALL NOT EXCEED 47 AMPERES.

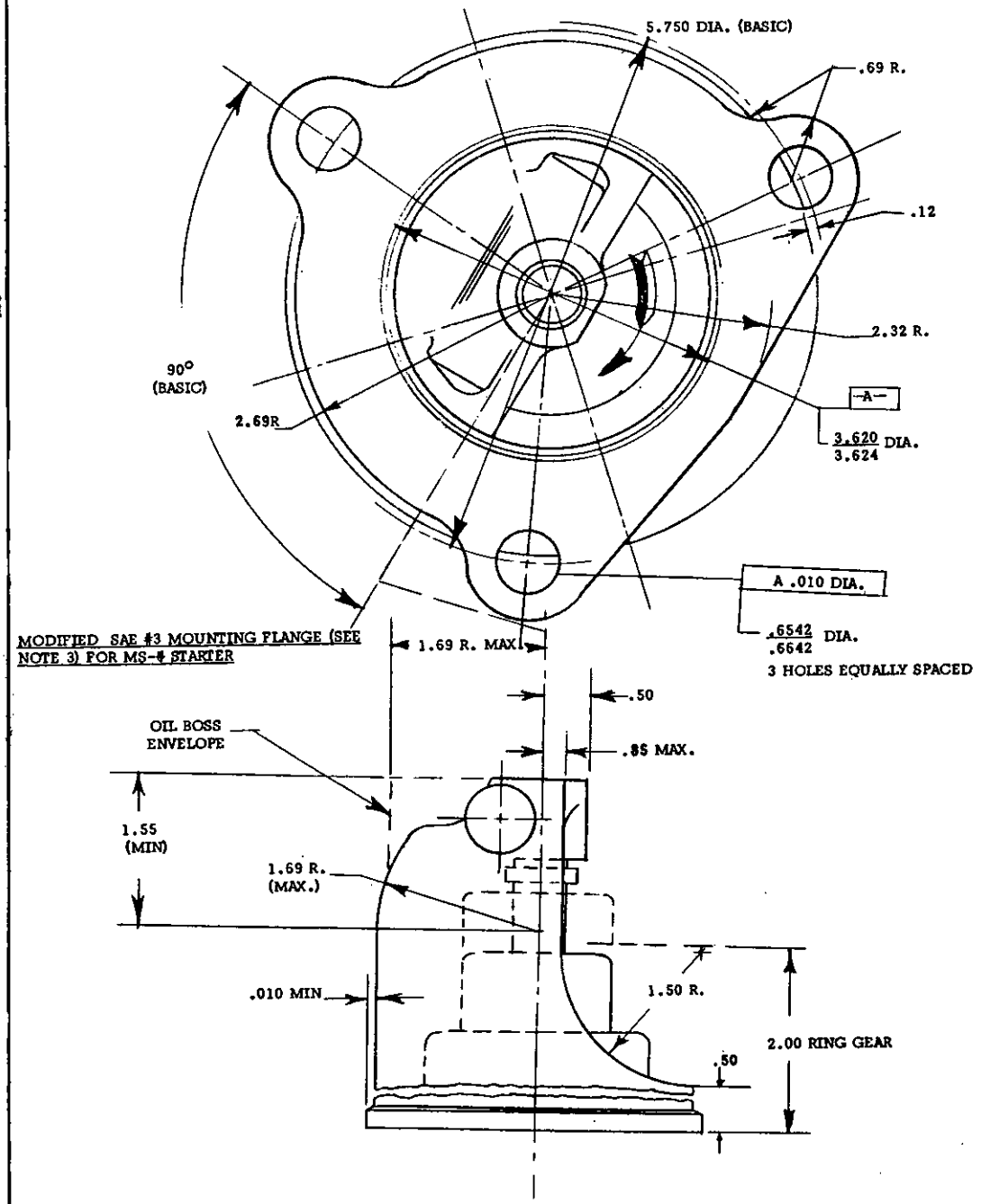
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P.A. ARMY-ME Other Cust AIR FORCE-82	TITLE STARTER, ENGINE, ELECTRICAL, 24-VOLT, D.C., 5-1/8-INCH FRAME DIAMETER, SOLENOID-ACTUATED, LIGHT-DUTY, SHORT AND LONG FRAME	MILITARY STANDARD MS 53010
PROCUREMENT SPECIFICATION MIL-S-3785	SUPERSEDES:	SHEET 7

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